

Structured Belief Revision and the Dynamics of Paradigm Change

A visual companion: paradigms as belief networks, revision as message passing

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Abstract

Belief revision in scientific communities exhibits a characteristic asymmetry. Peripheral commitments drift gradually under accumulating evidence, while central commitments resist revision long past the point where evidence favours alternatives, and when they do change, they tend to change quickly. We study this phenomenon within an active inference framework, recasting belief updating as motivated inference subject to a cost of changing one’s mind. Drawing on Lakatos and Kuhn, we represent an agent’s theoretical commitments as a structured network of interdependent beliefs rather than as a single latent variable. Each belief is a node that exchanges messages with its neighbours in both directions; a core of central commitments, on which much else depends, is the most costly to revise and the most resistant. Agents are embedded in a communication network through which they exchange their beliefs over the structure. This note is a figure-driven exposition of the construction, of the dynamics it is built to exhibit—the slow erosion of the belt and the rapid, percolation-like reorganisation of the core that we read as the signature of a paradigm shift—and of the question a structured belief raises that a scalar one does not: once an agent changes its mind, what exactly does it transmit?

1 From a scalar paradigm to a belief network

In the two-timescale account a paradigm is a single slow latent θ , and theory-ladenness is the dependence of the likelihood $p(o \mid s, \theta)$ on its value. That construction has no *internal* structure, so it cannot say why a paradigm shift looks the way it does: gradual at the edges, sudden at the centre. We give the paradigm structure. An agent’s theoretical commitments are a network of interdependent beliefs $\{x_i\}$. A subset of central commitments forms a densely connected *core* (A, B, C, D); the remaining nodes are a *belt* of more peripheral beliefs that carry the network’s contact with observation. The core is a designated part of the model’s update setup—the beliefs an agent treats as load-bearing—not a partition read off the graph. Each agent holds the same network; what differs between agents is the belief state over it, and agents communicate those states across a trust channel (Figure 1).

2 The network and its updates: message passing

Belief revision on the network is message passing. Each node takes input from its neighbours and sends output back to them, so every link carries two directed messages, one each way (Figure 2). A node’s belief is revised by combining the evidence it receives with the messages arriving from its neighbours; iterating this propagates a local change through the whole structure. The core is simply the set of densely connected central commitments through which most of these messages must pass.

3 Conservatism in the update setup

Each node is revised by the motivated, tempered rule $q'_i \propto q_{0,i} L_i^{\alpha/\lambda_i} e^{u_i/\lambda_i}$ (Hyland and Albarracín, 2025), where λ_i is the conservatism of node i : the cost of moving its belief. The core/belt difference

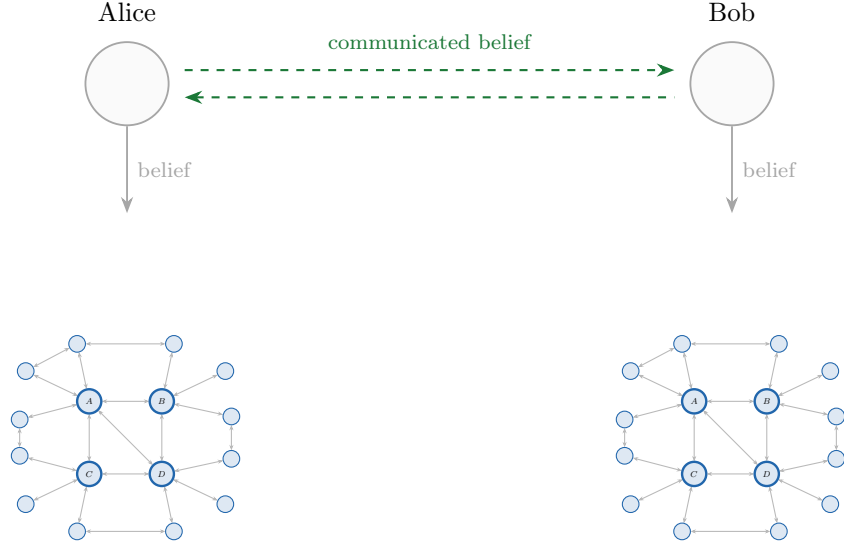


Figure 1: **The setup.** Two agents each carry a belief over the *same* structured network of commitments and exchange those beliefs across a trust channel (green, dashed). Within each network the four central nodes A, B, C, D (large) form the core; the surrounding nodes (small) are the peripheral belt that touches observation. Links inside a network carry messages in both directions (Figure 2).

in this cost is part of the update setup for the network, not a quantity read off the graph. Central commitments are assigned higher λ_i because much of the network’s interpretation rests on them—moving a core belief forces a great deal of re-reading downstream—while peripheral beliefs are cheap to revise (Figure 3). The graded shading is this assigned conservatism, the same gradient that will make the belt move before the core.

4 Belief-utility concentrates on the core

Conservatism λ_i is a *symmetric* stiffness—it resists motion in any direction. The affective belief-utility u_i is *directional*—it pulls the belief toward a preferred value regardless of evidence. The two are distinct costs that look alike in a single update but separate when evidence is sparse or when prior and preference disagree. Like conservatism, u_i is naturally concentrated on the core: agents have no feelings about which peripheral bookkeeping is correct, but strong commitments about the central claims (Figure 4). Conservatism gives convergence at varying speeds; a supra-threshold core u_i is what can produce permanent lock-in even as evidence accumulates.

5 Evidence enters the belt and propagates inward

An observation is evidentially connected to peripheral nodes. It updates them near-Bayesianly, because their λ_i is low, and message passing then carries the implication inward. As it advances toward the core the conservatism it meets rises, so the update slows (Figure 5). This is the gradient that produces the dynamics of the next section: evidence diffuses inward but is increasingly resisted, and the core moves only once enough of the belt has been re-interpreted.

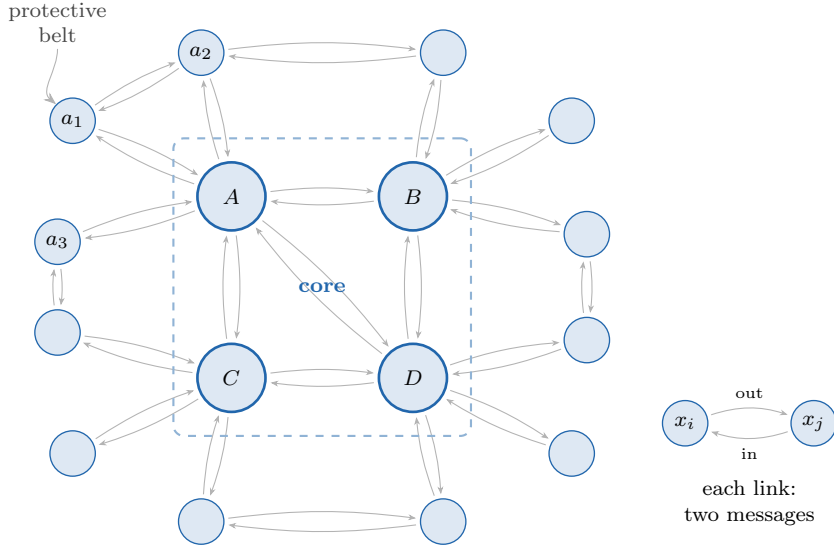


Figure 2: **The network and message passing.** The same network, core (A, B, C, D) boxed. Every link carries two directed messages—input from a neighbour and output to it (inset)—so revision is bidirectional propagation, not a one-way flow. The core is the densely connected centre through which most messages route; the belt is the network’s sensory surface.

6 The paradigm shift as a percolation cascade

Run the dynamics and a characteristic signature appears. For a long time only the belt moves: peripheral nodes flip to the new paradigm one by one as evidence accumulates, while the core holds. Each flip removes some of the inferential support the core relied on. Once enough of the belt has been hollowed out, the core can no longer be sustained against the evidence, and it gives way *together*, quickly—a percolation-like cascade (Figure 6). The macroscopic appearance is exactly Kuhn’s: long stability with visible activity only at the edges, then a sudden reorganisation. The discrete- θ model could not produce this, because it had no internal structure for a transition to propagate through.

7 Communicating a changed mind

Suppose the cascade runs in Alice: her belief over the network has moved, core and belt, to the new paradigm. She must now convey this to Bob across the trust channel. With a scalar θ the message was simply a posterior over K values. With a structured belief the message is a structured object, and it is no longer obvious *what* should be sent (Figure ??). She could transmit only the node marginals $\{q(x_i)\}$ —cheap, but discarding the correlations that made her change coherent; the full joint $q(x_1, \dots, x_n)$ —exact, and for a small network entirely feasible to send; or a set of factor fragments, conditional bundles such as “if A then B because C ”—the form closest to how scientific claims are actually communicated. The choice fixes how much of Alice’s reorganisation survives in Bob, and so how the cascade spreads through the population. We take it as an open modelling question rather than settling it here; for a small network the joint is the natural default, with fragments the more realistic refinement.

8 The macroscopic signature

Aggregating over the population gives the order parameters to measure. The mean peripheral belief drifts down smoothly as evidence converts the belt; the mean core belief stays near its

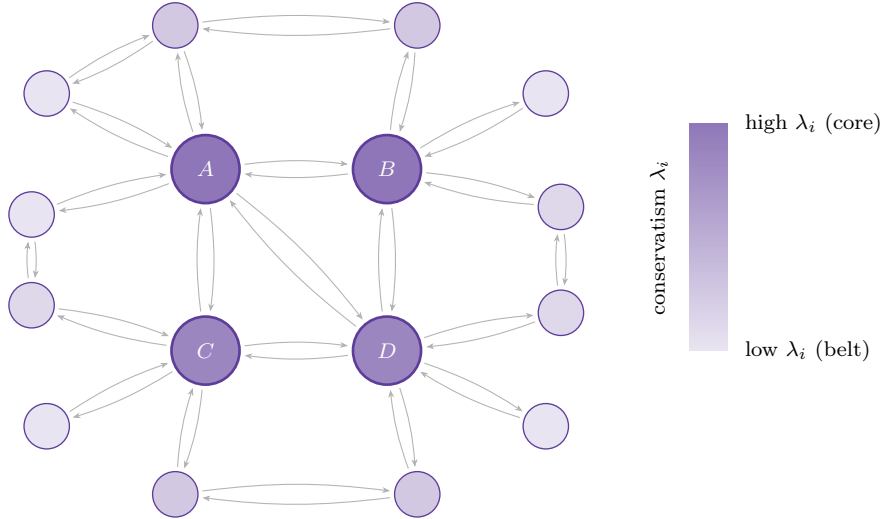


Figure 3: **Conservatism in the update setup.** Shade encodes λ_i , the cost of revising node i . Central commitments (dark) are assigned higher conservatism than peripheral ones (light), because much of the network’s interpretation rests on them. This assignment is part of how the network is set up to update; it is not derived from the directed structure of the graph.

prior, then drops sharply once the percolation threshold is crossed (Figure 8a). Across the coupled population, with one-round transmission delays, the approach to the new consensus is a damped oscillation whose damping is set by the distribution of conservatism $\{\lambda_i\}$ and the spectral structure of the trust matrix (Figure 8b). The curves are illustrative schematics of the target phenomenology, not simulation output.

Outlook

Structuring the paradigm turns the phase-shift signature into a consequence rather than an assumption: it is a percolation cascade through a network whose core is held by the conservatism of the update setup. It also raises a question the scalar model could not pose—once a mind has changed, what is communicated: marginals, the joint, or inferential fragments. The same machinery is invariant under a change of who controls attention over the network: in science, attention is shaped by the agent and its community; in algorithmically curated settings it is shaped externally, and beliefs propagate by salience rather than by trust—the formal object is the same, the generator of the attention distribution is not.

References

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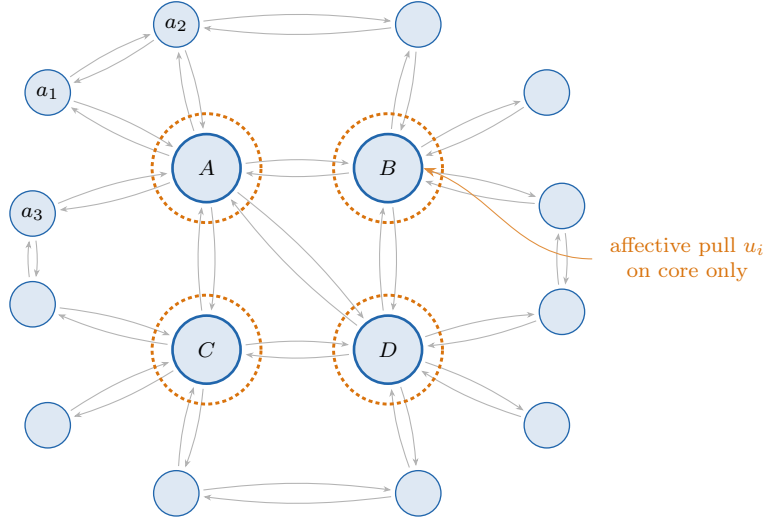


Figure 4: **Belief-utility on the core.** Orange haloes mark nodes with nonzero affective utility u_i : a preference for a particular value of the central commitments, independent of fit. The belt is affectively neutral. The update $q'_i \propto q_{0,i} L_i^{\alpha/\lambda_i} e^{u_i/\lambda_i}$ runs node-wise; only the core carries the e^{u_i/λ_i} tilt.

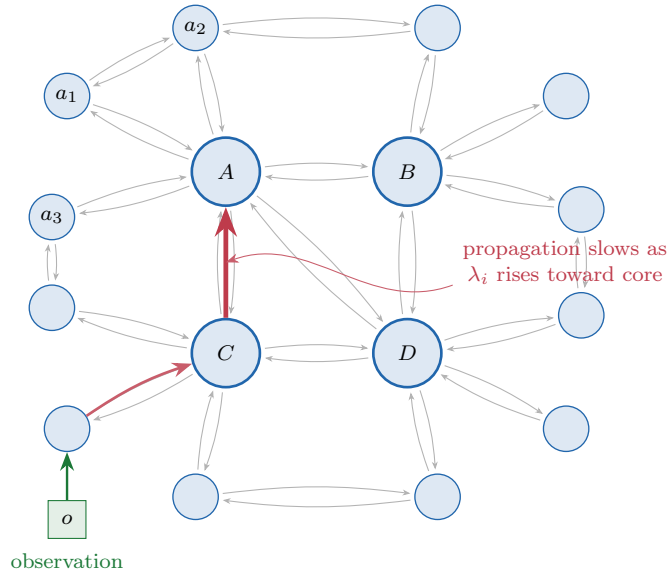


Figure 5: **Inward propagation against rising conservatism.** An observation o (green) enters at a leaf and flips it cheaply. Its implication propagates inward (red), but each step toward the core meets higher λ_i (thickening arrow = greater resistance). Peripheral nodes track the data; the core lags.

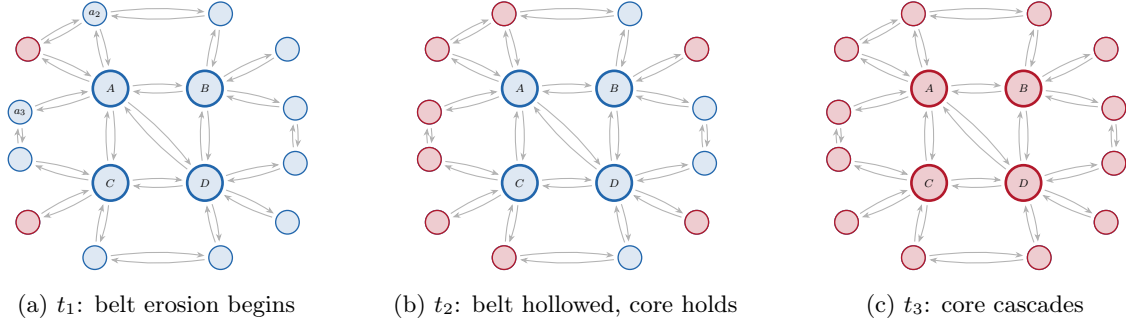


Figure 6: **Percolation cascade = phase shift.** Blue is the old paradigm, red the new. (a) A few peripheral nodes flip cheaply. (b) Evidence has converted most of the belt, but the high- λ core still resists—the long quiet period, with activity only at the edges. (c) With the belt’s support gone the core can no longer be defended and flips as a block. The transition is gradual at the periphery and sudden at the centre.

Alice (changed her mind)

Bob (still to be convinced)

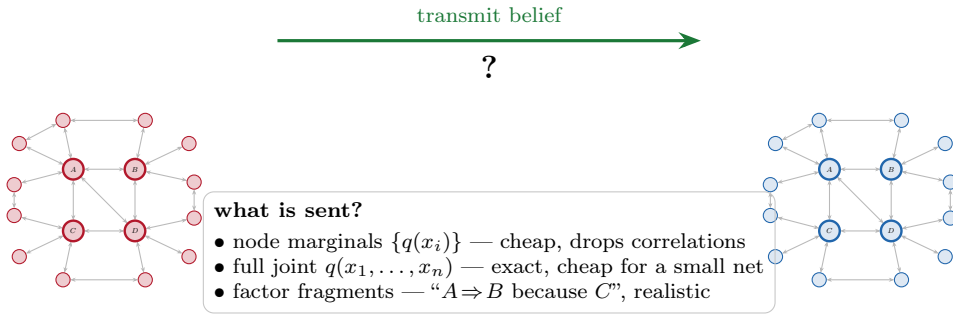
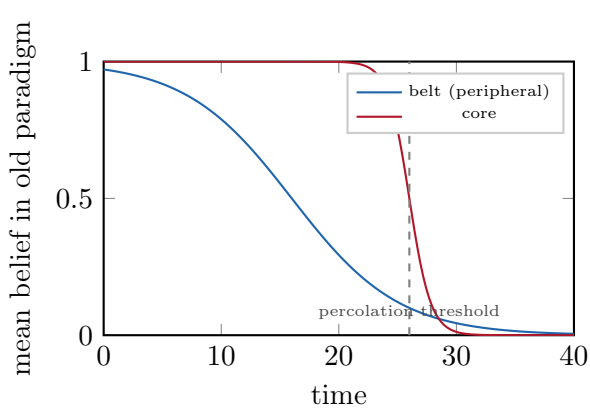
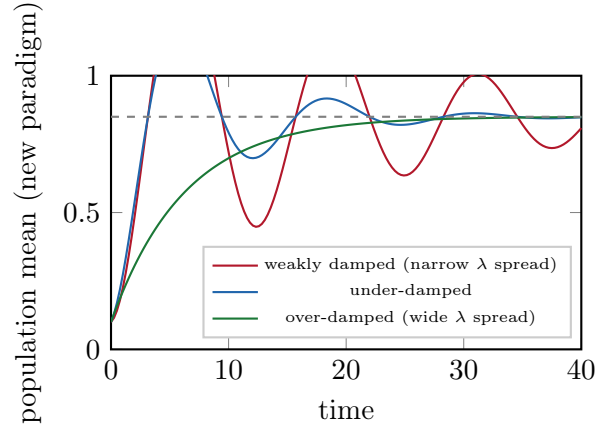


Figure 7: **Communicating a changed mind.** After the cascade, Alice’s whole network has moved to the new paradigm (red); Bob’s has not (blue). To convey her new belief she must choose what to transmit: node marginals (cheap, lossy), the full joint (exact, and feasible for a small network), or factor fragments (claims with their inferential support, closest to real scientific communication). The choice determines how much of her reorganisation survives in Bob—an open question the scalar model never had to face.



(a) gradual belt drift, sudden core flip



(b) approach to consensus vs. $\{\lambda_i\}$

Figure 8: Macroscopic signature (schematic; not simulation output). (a) The mean peripheral belief drifts smoothly while the mean core belief holds, then drops sharply at the percolation threshold (dashed)—the order-parameter form of a Kuhnian phase shift. (b) Coupled through the trust network with transmission delay, the population approaches the new consensus as a damped oscillation whose regime is governed by the distribution of conservatism and the spectrum of the trust matrix.